

Отчёт по лабораторной работе №4

Продвинутое использование git

Арсланов Довлетгелди

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1 Цель работы

Получение навыков правильной работы с репозиториями git.

2 Теоретические сведения

- Gitflow Workflow опубликована и популяризована Винсентом Дриссенем.
- Gitflow Workflow предполагает выстраивание строгой модели ветвления с учётом выпуска проекта.
- Данная модель отлично подходит для организации рабочего процесса на основе релизов.
- Работа по модели Gitflow включает создание отдельной ветки для исправлений ошибок в рабочей среде.
- Последовательность действий при работе по модели Gitflow:
- Из ветки master создаётся ветка develop.
- Из ветки develop создаётся ветка release.
- Из ветки develop создаются ветки feature.
- Когда работа над веткой feature завершена, она сливается с веткой develop.
- Когда работа над веткой релиза release завершена, она сливается в ветки develop и master.
- Если в master обнаружена проблема, из master создаётся ветка hotfix.

- Когда работа над веткой исправления hotfix завершена, она сливается в ветки develop и master.

3 Выполнение лабораторной работы

3.1 Работа с тестовым репозиторием

Для работы с Node.js добавим каталог с исполняемыми файлами, устанавливаемыми yarn, в переменную PATH.

```

darslanov@darslanov:~$ pnpm setup
Appended new lines to /home/darslanov/.bashrc

Next configuration changes were made:
export PNPM_HOME="/home/darslanov/.local/share/pnpm"
case ":$PATH:" in
  *:$PNPM_HOME:*) ;;
  *) export PATH="$PNPM_HOME:$PATH" ;;
esac

To start using pnpm, run:
source /home/darslanov/.bashrc
darslanov@darslanov:~$
^[[200~source ~/.bashrc
^[[201~darslanov@darslasource ~/.bashrc.bashrc
darslanov@darslanov:~$ source ~/.bashrc
darslanov@darslanov:~$ █

```

Рисунок 3.1: Node.js

Программа commitizen используется для помощи в форматировании коммитов. При этом устанавливается скрипт git-cz, который мы и будем использовать для коммитов.

```

darslanov@darslanov:~$ pnpm add -g commitizen

Update available! 10.27.0 → 11.25.0.
Changelog: https://pnpm.io/v/11.25.0
To update, run: pnpm self-update

WARN GET https://registry.npmjs.org/cz-conventional-changelog error (ECONNRESET). Will retry in 10 seconds. 2 retries left.
WARN GET https://registry.npmjs.org/minimist error (ECONNRESET). Will retry in 10 seconds. 2 retries left.
WARN Request took 31558ms: https://registry.npmjs.org/cz-conventional-changelog
WARN Request took 31565ms: https://registry.npmjs.org/minimist
WARN GET https://registry.npmjs.org/once error (ECONNRESET). Will retry in 10 seconds. 2 retries left.
WARN GET https://registry.npmjs.org/once error (ECONNRESET). Will retry in 1 minute. 1 retries left.
WARN Request took 108240ms: https://registry.npmjs.org/once
WARN 2 deprecated subdependencies found: glob@7.2.3, inflight@1.0.6
Packages: +148
+-----+
Downloading @typescript/typescript-linux-x64@7.0.2: 9.67 MB/9.67 MB, done
WARN GET https://registry.npmjs.org/balanced-match/-/balanced-match-1.0.2.tgz error (ECONNRESET). Will retry in 10 seconds. 2 retries left.
Progress: resolved 167, reused 0, downloaded 148, added 148, done

/home/darslanov/.local/share/pnpm/global/5:
+ commitizen 4.3.2

Done in 2m 49.6s using pnpm v10.27.0
darslanov@darslanov:~$

```

Рисунок 3.2: установка commitizen

Программа standard-changelog используется для помощи в создании логов.

```
darslanov@darslanov:~$ pnpm add -g standard-changelog
WARN 2 deprecated subdependencies found: glob@7.2.3, inflight@1.0.6
Packages: +14
+++++
Progress: resolved 181, reused 148, downloaded 14, added 14, done

/home/darslanov/.local/share/pnpm/global/5:
+ standard-changelog 8.1.0

Done in 1.4s using pnpm v10.27.0
darslanov@darslanov:~$
```

Рисунок 3.3: установка standard-changelog

Делаем первый коммит и выкладываем на github.

Необходимо заполнить несколько параметров пакета.

Таким образом, файл package.json приобретает вид:

```
1  
2   "name": "git-extended",  
3   "version": "1.0.0",  
4   "description": "Git repo for educational purposes",  
5   "main": "index.js",  
6   "repository": "git@github.com:d-arslanov/git-extended.git",  
7   "author": "d-arslanov",  
8   "license": "CC-BY-4.0",  
9   "config": {  
10     "commitizen": {  
11       "path": "cz-conventional-changelog"  
12     }  
13   }  
14
```

Рисунок 3.4: package.json

Добавим новые файлы.

Выполним коммит.

Отправим на github.

```

darslanov@darslanov:~/git-extended$ git add .
darslanov@darslanov:~/git-extended$ git cz
cz-cli@4.3.2, cz-conventional-changelog@3.3.0

? Select the type of change that you're committing: chore:  Other changes that don't modify src or
test files
? What is the scope of this change (e.g. component or file name): (press enter to skip)
? Write a short, imperative tense description of the change (max 93 chars):
(4) test
? Provide a longer description of the change: (press enter to skip)

? Are there any breaking changes? No
? Does this change affect any open issues? No
[master 2e6985e] chore: test
1 file changed, 14 insertions(+)
create mode 100644 package.json
darslanov@darslanov:~/git-extended$ git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 478 bytes | 478.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:d-arslanov/git-extended.git
132eca4..2e6985e master -> master
darslanov@darslanov:~/git-extended$

```

Рисунок 3.5: Отправка

Инициализируем git-flow

Проверьте, что Вы на ветке develop

Загрузите весь репозиторий в хранилище

```

darslanov@darslanov:~/git-extended$ git flow init

Which branch should be used for bringing forth production releases?
- master
Branch name for production releases: [master]
Branch name for "next release" development: [develop]

How to name your supporting branch prefixes?
Feature branches? [feature/]
Bugfix branches? [bugfix/]
Release branches? [release/]
Hotfix branches? [hotfix/]
Support branches? [support/]
Version tag prefix? []
Hooks and filters directory? [/home/darslanov/git-extended/.git/hooks]
darslanov@darslanov:~/git-extended$ git branch
* develop
  master
darslanov@darslanov:~/git-extended$ git push --all
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'develop' on GitHub by visiting:
remote:   https://github.com/d-arslanov/git-extended/pull/new/develop
remote:
To github.com:d-arslanov/git-extended.git
* [new branch]    develop -> develop
darslanov@darslanov:~/git-extended$

```

Рисунок 3.6: Инициализация и загрузка изменений

Установите внешнюю ветку как вышестоящую для этой ветки

Создадим релиз с версией 1.0.0

Создадим журнал изменений

Добавим журнал изменений в индекс

Зальём релизную ветку в основную ветку

```

Summary of actions:
- A new branch 'release/1.0.0' was created, based on 'develop'
- You are now on branch 'release/1.0.0'

Follow-up actions:
- Bump the version number now!
- Start committing last-minute fixes in preparing your release
- When done, run:

    git flow release finish '1.0.0'

darslanov@darslanov:~/git-extended$ standard-changelog --first-release
darslanov@darslanov:~/git-extended$ git add CHANGELOG.md
darslanov@darslanov:~/git-extended$ git commit -am 'chore(site): add changelog'
[release/1.0.0 cfb57c4] chore(site): add changelog
1 file changed, 1 insertion(+)
create mode 100644 CHANGELOG.md
darslanov@darslanov:~/git-extended$ git flow release finish 1.0.0
Switched to branch 'master'
Your branch is up to date with 'origin/master'.
Merge made by the 'ort' strategy.
CHANGELOG.md | 1 +
1 file changed, 1 insertion(+)
create mode 100644 CHANGELOG.md
Already on 'master'
Your branch is ahead of 'origin/master' by 2 commits.
(use "git push" to publish your local commits)
Switched to branch 'develop'
Your branch is up to date with 'origin/develop'.
Merge made by the 'ort' strategy.
CHANGELOG.md | 1 +

```

Рисунок 3.7: Завершение релиза

Отправим данные на github

```
darslanov@darslanov:~/git-extended$ git push --all
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 547 bytes | 547.00 KiB/s, done.
Total 5 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), done.
To github.com:d-arslanov/git-extended.git
  2e6985e..11fe83f  develop -> develop
  2e6985e..af4044d  master -> master
darslanov@darslanov:~/git-extended$ git push --tags
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 155 bytes | 155.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:d-arslanov/git-extended.git
 * [new tag]          1.0.0 -> 1.0.0
darslanov@darslanov:~/git-extended$ gh release create v1.0.0 -F CHANGELOG.md
https://github.com/d-arslanov/git-extended/releases/tag/v1.0.0
darslanov@darslanov:~/git-extended$
```

Рисунок 3.8: Отправка

Создадим ветку для новой функциональности По окончании разработки новой функциональности следующим шагом следует объединить ветку `feature_branch` с `develop`:

```
darslanov@darslanov:~/git-extended$ git flow feature start feature_branch
Switched to a new branch 'feature/feature_branch'

Summary of actions:
- A new branch 'feature/feature_branch' was created, based on 'develop'
- You are now on branch 'feature/feature_branch'

Now, start committing on your feature. When done, use:

    git flow feature finish feature_branch

darslanov@darslanov:~/git-extended$ touch test2
darslanov@darslanov:~/git-extended$ git flow feature finish feature_branch
Switched to branch 'develop'
Your branch is up to date with 'origin/develop'.
Already up to date.
Deleted branch feature/feature_branch (was 11fe83f).

Summary of actions:
- The feature branch 'feature/feature_branch' was merged into 'develop'
- Feature branch 'feature/feature_branch' has been locally deleted
- You are now on branch 'develop'

darslanov@darslanov:~/git-extended$
```

Рисунок 3.9: Объединение веток

Создадим релиз с версией 1.2.3

Обновите номер версии в файле package.json. Установите её в 1.2.3

Создадим журнал изменений

Добавим журнал изменений в индекс

Зальём релизную ветку в основную ветку

```

Summary of actions:
- Release branch 'release/1.2.3' has been merged into 'master'
- The release was tagged '1.2.3'
- Release tag '1.2.3' has been back-merged into 'develop'
- Release branch 'release/1.2.3' has been locally deleted
- You are now on branch 'develop'

darslanov@darslanov:~/git-extended$ git push --all
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 8 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 747 bytes | 747.00 KiB/s, done.
Total 6 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 1 local object.
To github.com:d-arslanov/git-extended.git
    11fe03f..2e113f7  develop -> develop
    af4044d..b7a77f1  master -> master
darslanov@darslanov:~/git-extended$ git push --tags
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 155 bytes | 155.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:d-arslanov/git-extended.git
    * [new tag]         1.2.3 -> 1.2.3
darslanov@darslanov:~/git-extended$ gh release create v1.2.3 -F CHANGELOG.md
https://github.com/d-arslanov/git-extended/releases/tag/v1.2.3
darslanov@darslanov:~/git-extended$

```

Рисунок 3.10: Завершение релиза

3.2 Подготовка рабочего репозитория

```
darslanov@darslanov:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$ gedit package.json
darslanov@darslanov:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$ git add .
darslanov@darslanov:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$ git cz
cz-cli@4.3.2, cz-conventional-changelog@3.3.0

? Select the type of change that you're committing: chore: Other changes that don't modify src or test files
? What is the scope of this change (e.g. component or file name): (press enter to skip)
? Write a short, imperative tense description of the change (max 93 chars):
(4) test
? Provide a longer description of the change: (press enter to skip)

? Are there any breaking changes? No
? Does this change affect any open issues? No
[master 257eb3d] chore: test
1 file changed, 11 insertions(+), 11 deletions(-)
darslanov@darslanov:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$ git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 474 bytes | 474.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:d-arslanov/2026-1--study--os-intro.git
fbf79ab..257eb3d master -> master
darslanov@darslanov:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$
```

Рисунок 3.11: package.json и коммит

```

Summary of actions:
- Release branch 'release/1.0.0' has been merged into 'master'
- The release was tagged '1.0.0'
- Release tag '1.0.0' has been back-merged into 'develop'
- Release branch 'release/1.0.0' has been locally deleted
- You are now on branch 'develop'

darslanov@darslanov:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$ git push --all

Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 8 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 719 bytes | 719.00 KiB/s, done.
Total 5 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 1 local object.
To github.com:d-arslanov/2026-1--study--os-intro.git
   257eb3d..4a624e3  develop -> develop
   257eb3d..c14e744  master -> master
darslanov@darslanov:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$ git push --tags
s
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 155 bytes | 155.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:d-arslanov/2026-1--study--os-intro.git
 * [new tag]         1.0.0 -> 1.0.0
darslanov@darslanov:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$

```

Рисунок 3.12: Завершение релиза

4 Вывод

Мы приобрели практические навыки взаимодействия с дополнительными функциями гитхаб.